

- Hybrid apps

Native mobile/smartphone apps

The term 'native mobile app' refers to an application that is developed in the base programming language. A native Android app is always developed using the Android SDK. Similarly, for Apple based mobile phones, smartphone applications are developed using iOS as the specific programming language for that platform.

The following are the key limitations when developing native apps:

- A separate team for each platform is required for troubleshooting and making changes in the app.
- There are huge development costs associated with each development environment.
- Multiple apps are required for each platform because of compatibility issues.
- No code-sharing is possible in multiple versions. There is a need to create separate code for every new version.
- The development of the application is slow.
- Testing takes longer for multiple flavours or versions.

Cross-platform mobile/smartphone apps

A single source code is used in the case of cross-platform mobile app development. It works on the strategy of 'Write once, deploy anywhere'. Using such platforms, the source code is written once and the final executable application can be deployed on any platform including Android, Windows, Tizen, Ubuntu, Firefox, iOS or any other, without any changes in the code. This approach costs much less because development and troubleshooting is done by one team rather than multiple teams for each platform.

The tools and technologies for the development of cross-platform mobile apps include Cordova, Codename One, PhoneGap, Appcelerator, Sencha Touch, Monocross, Kony, NativeScript, RhoMobile, Xamarin and many others.

The following are the advantages of using cross-platform mobile apps:

- Code-sharing is possible in multiple versions and multiple platforms.
- A single developer or team is sufficient to develop apps for all platforms.
- Fast performance on the user interface.
- Easy to program in HTML, JavaScript and CSS without any complex coding.
- Easy version control in multiple flavours of the mobile app.

Hybrid mobile/smartphone apps

Hybrid mobile apps combine the features of native and Web apps. They make use of embedded components with greater focus on Web applications. Traditionally, the hybrid apps integrate the code of HTML, CSS and JavaScript.

The platforms for the development of hybrid apps include

Cordova, Framework 7, Trigger, Ionic, Telerik, Mobile Angular UI, Onsen UI, Kendo UI, Quasar, NativeScript, React Native, Sencha UI and many others.

The key advantages of using hybrid mobile apps include the following:

- In-built libraries with enormous modules.
- Code-sharing for compatibility in multiple apps.
- Web as well as mobile compatibility.
- App updation can be done without the need to pull out the app from the store.
- Fast development and rapid updates are possible in the app for enhanced functionality.

The tremendous scope for cross-platform mobile apps

With the increasing number of mobile platforms, the development cost of smartphone apps is also increasing. Organisations need to create and deploy their apps for every type of mobile platform. As the cost of hiring developers for each platform is very high, there is a huge need for cross-platform mobile apps that can be deployed on any type of device, irrespective of whether it runs on Android, Windows, iOS, Symbian, Bada or any other OS. If organisations develop their applications for n number of individual platforms, the development costs increase n times

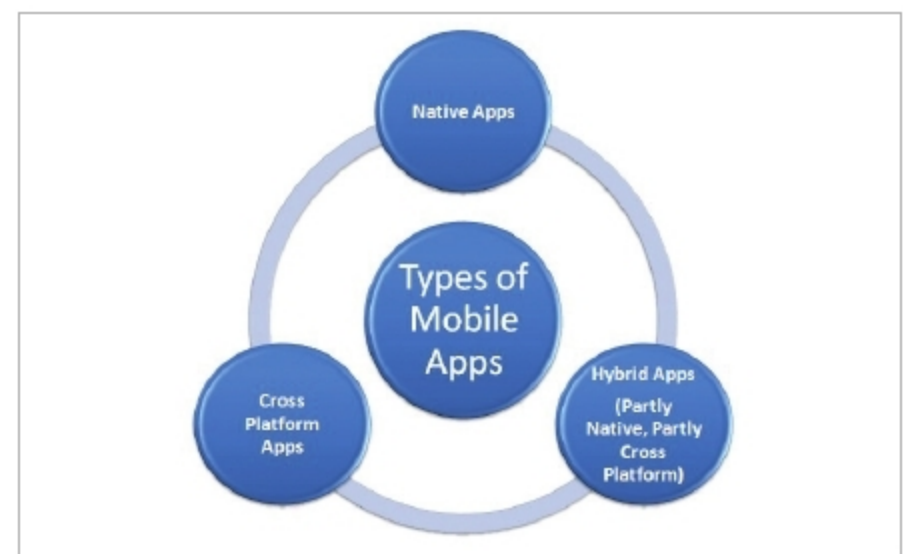


Figure 1: Categories of mobile apps

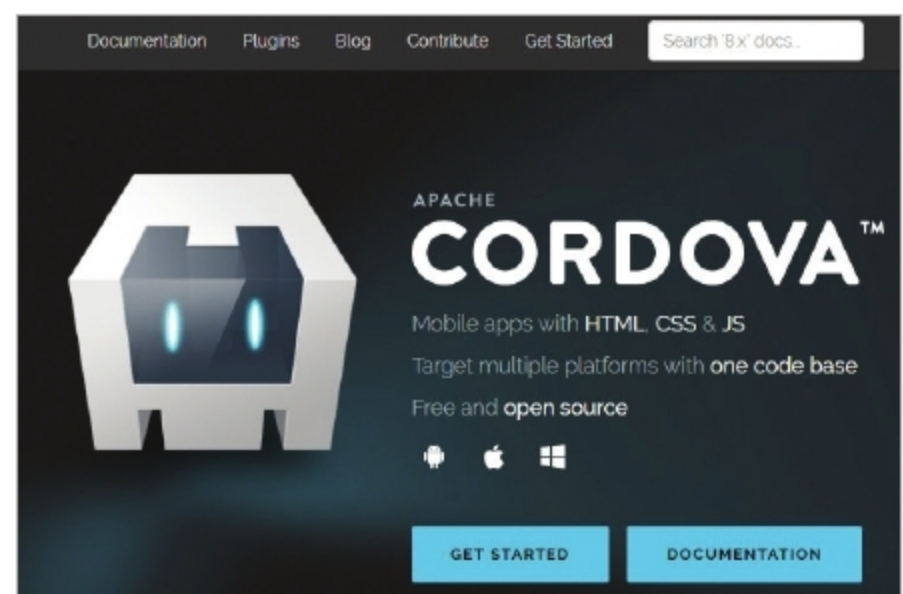


Figure 2: Official portal of Apache Cordova